

UK Electricity Consumption Analysis Dashboard

Year

All

Weekday

All

Month

All

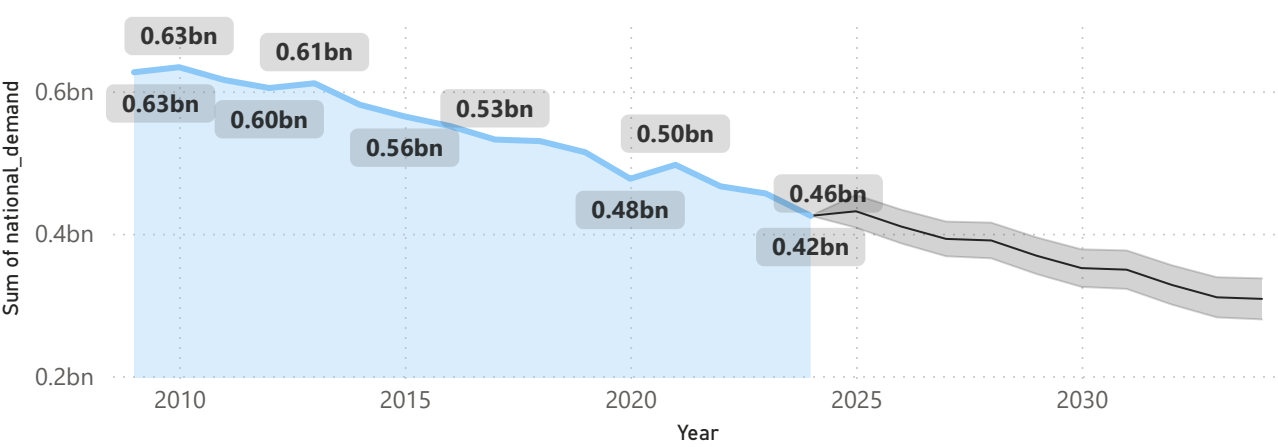
Is_holiday

All

Demand Heatmap (Weekday vs. Hour)

weekday	0	1	2	3	4	5	6
0	39874272	39191406	38268158	37394499	36973514	38763308	459
1	42145323	41256640	40218424	39289517	38777388	40478637	477
2	42441825	41490995	40410494	39430507	38904567	40608272	478
3	42700245	41707070	40592935	39606270	39060995	40732486	478
4	42695100	41617981	40455422	39410844	38817488	40367057	472
5	42162545	40741222	39221610	37892451	36892942	37064621	398
Total	292801468	285404152	277056430	269574608	264906965	273243515	3132

Total Energy Demand Over Time



31.19K

Avg Monthly Demand

599M

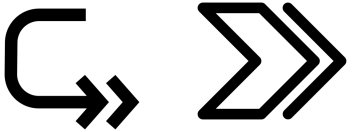
Total Renewable Generation

9bn

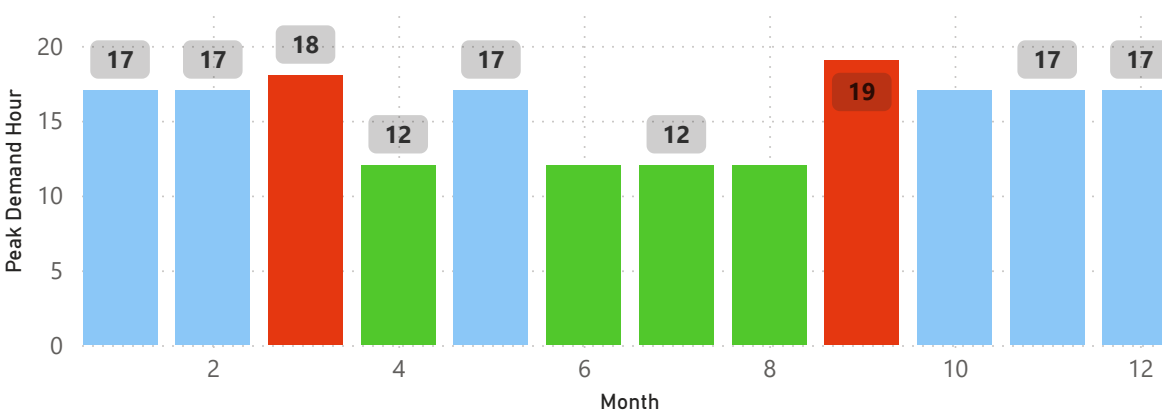
Total National Demand

0.07

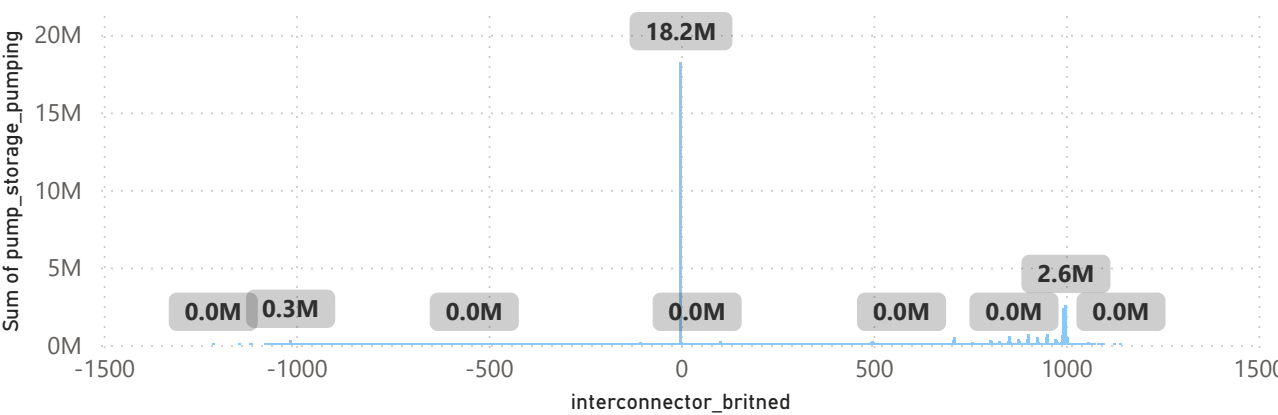
Renewable %



Peak Demand Hour by Month



Net Interconnector Energy Flow





Demand Analysis

Year



All



Weekday



All



Month



All



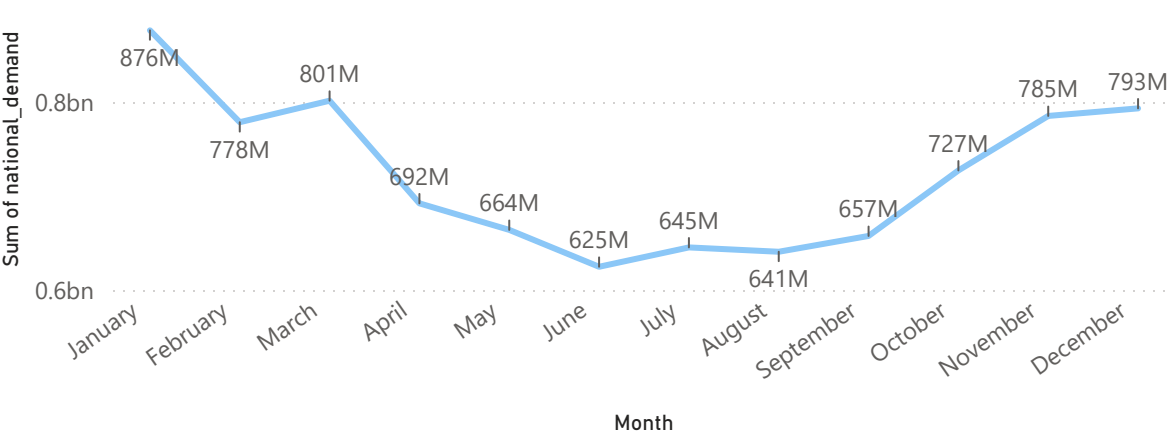
Is_holiday



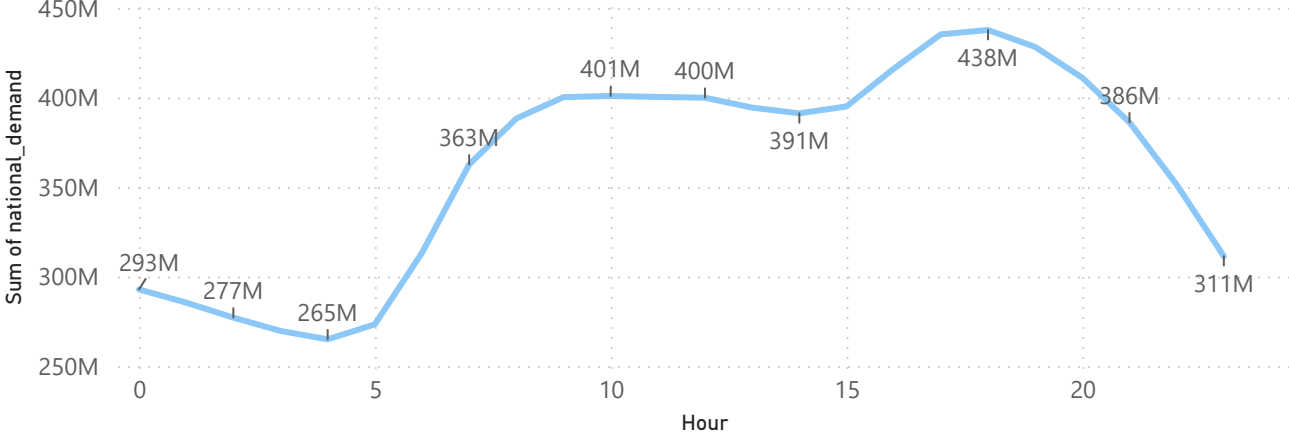
All



Monthly Demand Trends



Hourly Demand Pattern



31.19K

Avg Monthly Demand

599M

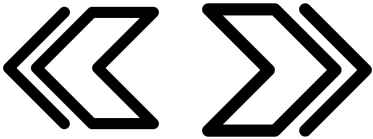
Total Renewable Generation

9bn

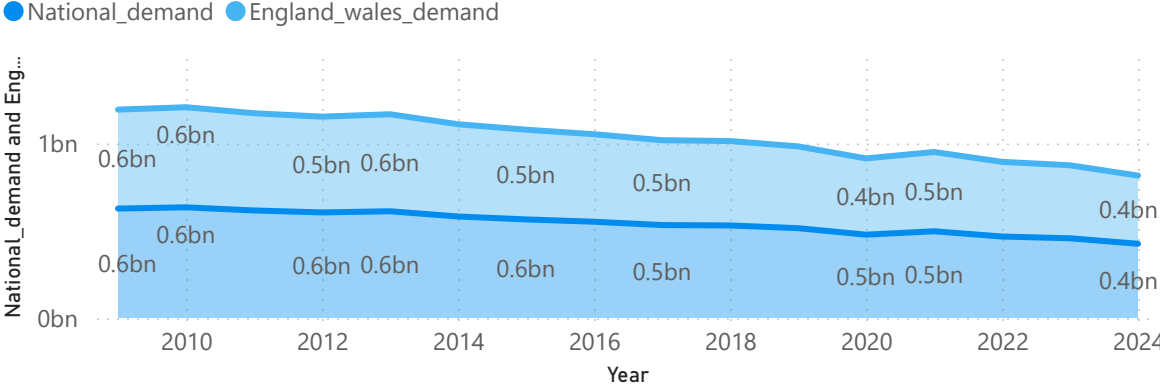
Total National Demand

0.07

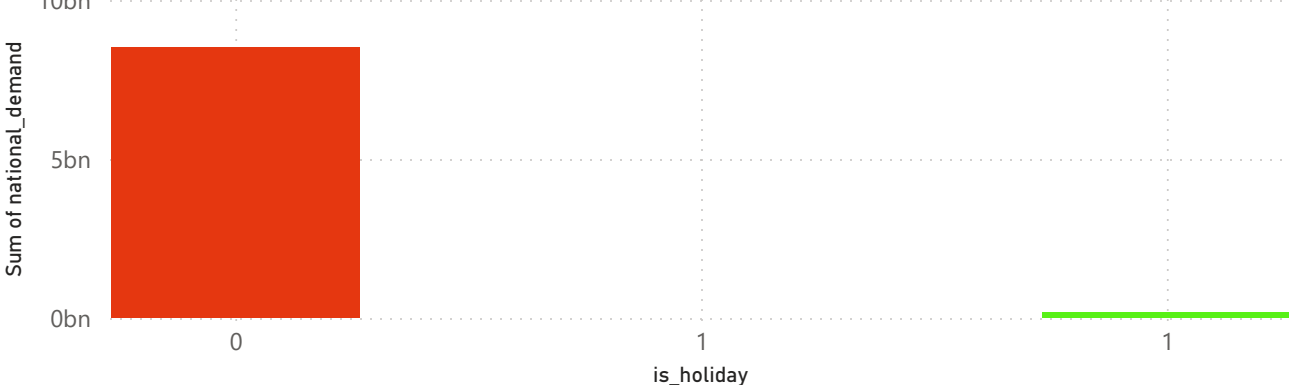
Renewable %



Demand Comparison: National vs. Regional



Holidays vs. Non-Holiday Demand





Renewable Energy Analysis

Year

All

Weekday

All

Month

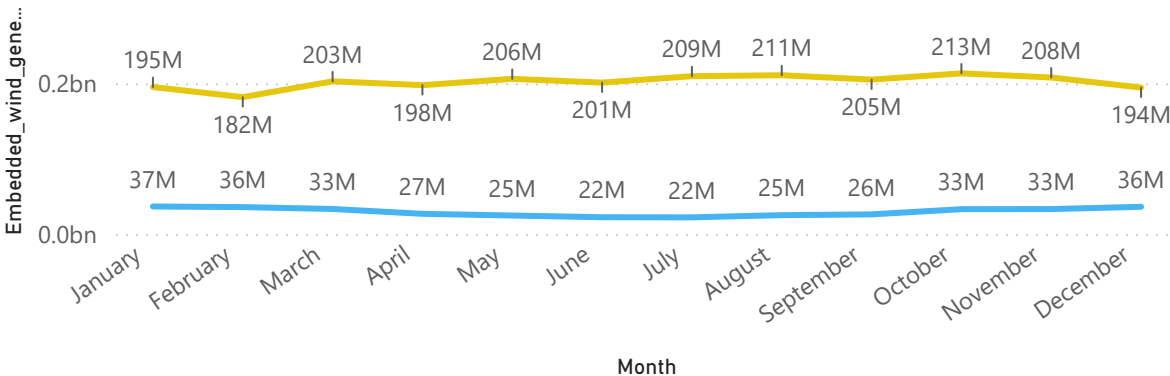
All

Is_holiday

All

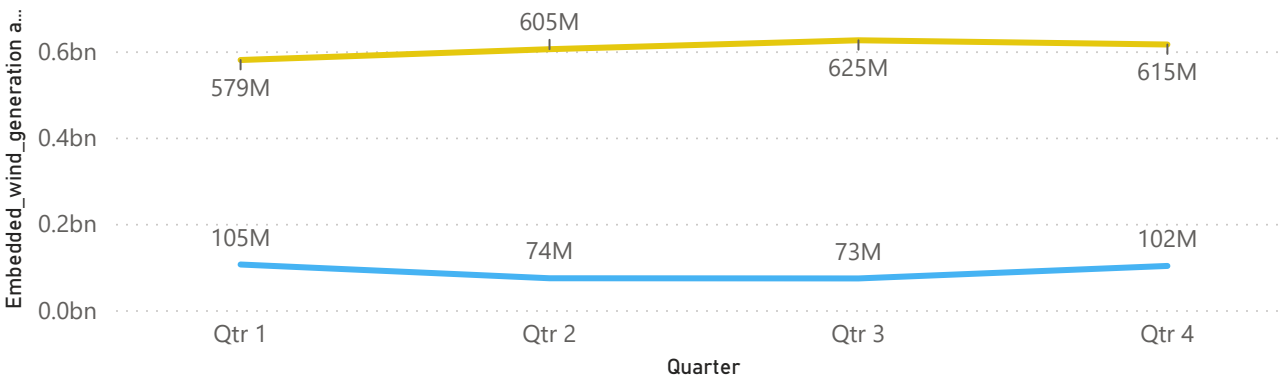
Solar & Wind Energy Trends

Embedded_wind_generation Embedded_solar_capacity



Solar & Wind Energy Trends

Embedded_wind_generation Embedded_solar_capacity



599M

Total Renewable Generation

8bn

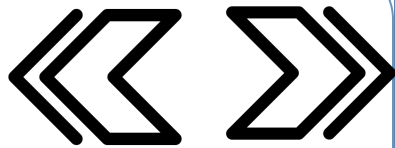
Total_Non_Renewable

6.90

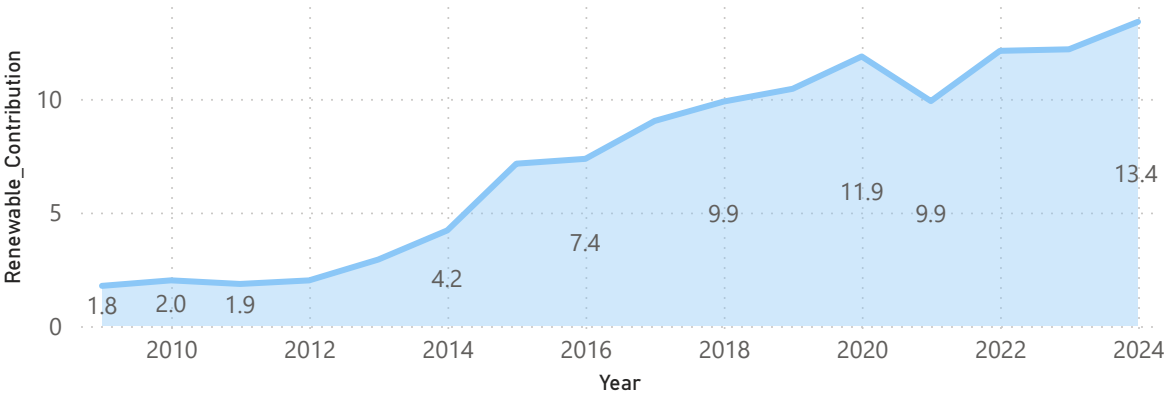
Renewable_Contribution

0.07

Renewable %

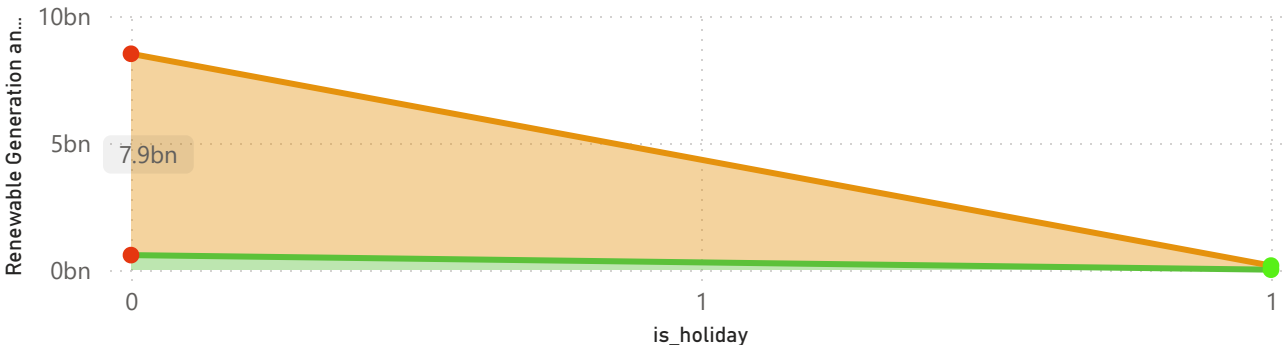


Renewable % Contribution Over Time



Renewable vs. Non-Renewable

Renewable Generation Non_Renewable



Interconnector Energy Flows

Year

All

Weekday

Month 

All 

Is_holiday

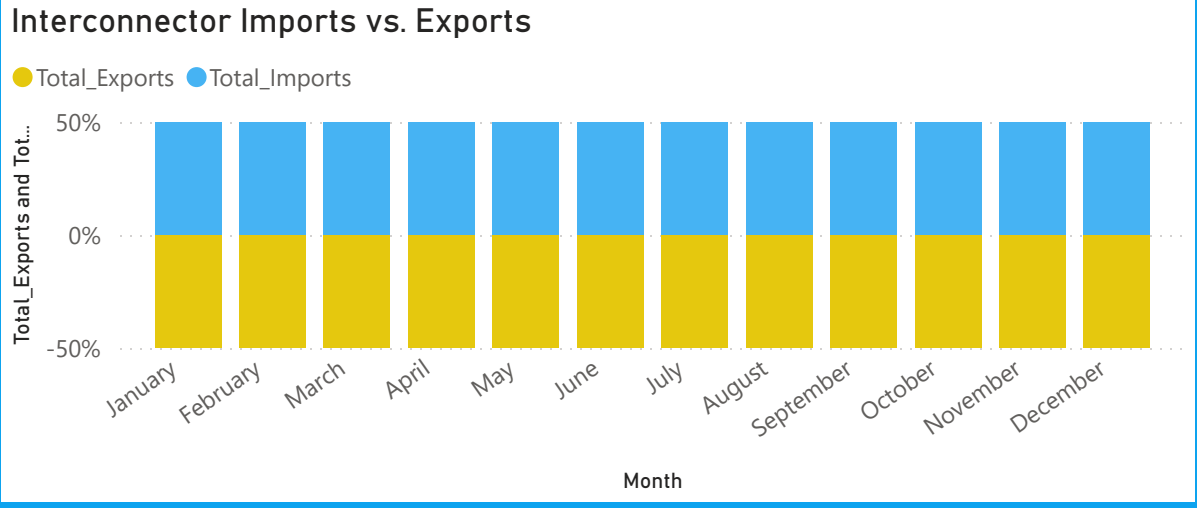
Interconnector Imports vs. Exports

Legend: Total_Exports (Yellow), Total_Imports (Blue)

Y-axis: Total_Exports and Tot... (ranging from -50% to 50%)

X-axis: Month (January to December)

Month	Total_Exports (%)	Total_Imports (%)
January	50	50
February	50	50
March	50	50
April	50	50
May	50	50
June	50	50
July	50	50
August	50	50
September	50	50
October	50	50
November	50	50
December	50	50



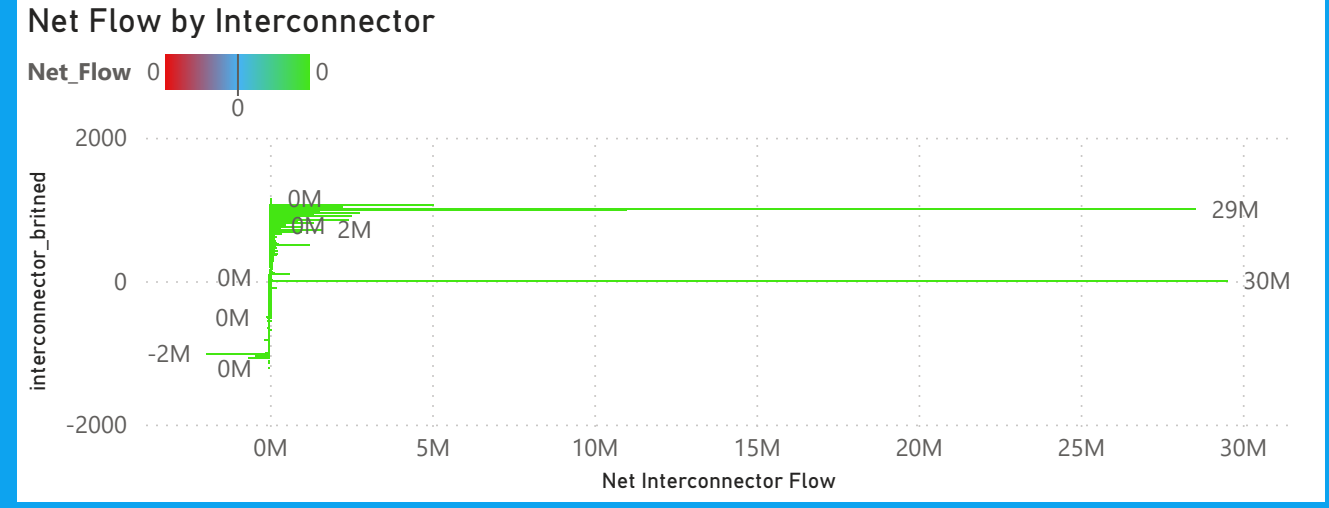
Net Flow by Interconnector

Net_Flow 0 0

interconnector_binned

Net Interconnector Flow

0M 2M 29M 30M



599M

Total Renewable Generation

599M

Total Renewable Generation

8bn

Total_Non_Renewable

8bn

Total_Non_Renewable

413M

Total_Imports

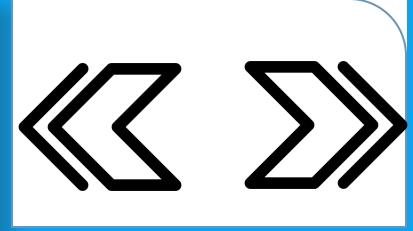
413M

Total_Imports

0.07
Renewable %

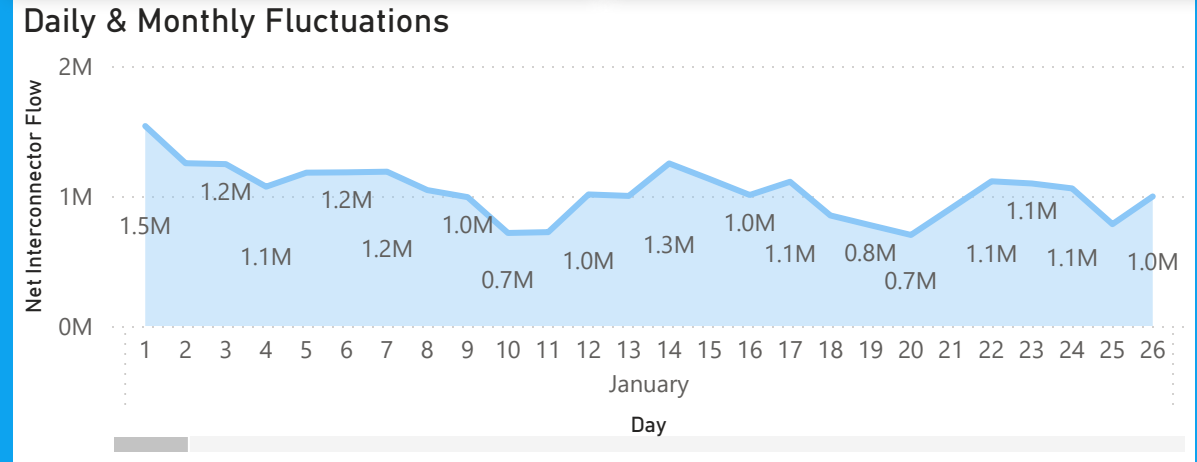
0.07

Renewable %



Daily & Monthly Fluctuations

Day	Net Interconnector Flow (M)
1	1.5M
2	1.2M
3	1.2M
4	1.1M
5	1.2M
6	1.2M
7	1.2M
8	1.0M
9	1.0M
10	0.7M
11	0.7M
12	1.0M
13	1.0M
14	1.3M
15	1.0M
16	1.0M
17	1.1M
18	0.8M
19	0.8M
20	0.7M
21	1.1M
22	1.1M
23	1.1M
24	1.1M
25	1.0M
26	1.0M

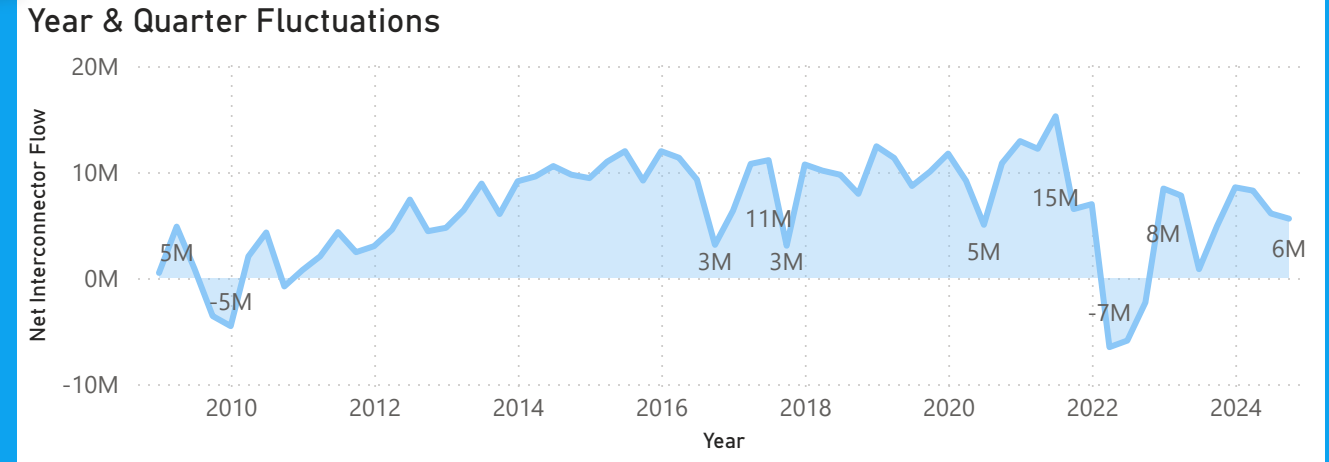


Year & Quarter Fluctuations

Net Interconnector Flow

Year

Year	Net Interconnector Flow (M)
2010	5M
2010	-5M
2017	3M
2017	11M
2018	3M
2020	5M
2021	15M
2022	-7M
2023	8M
2024	6M





Renewable Energy Analysis

Year

All

Weekday

All

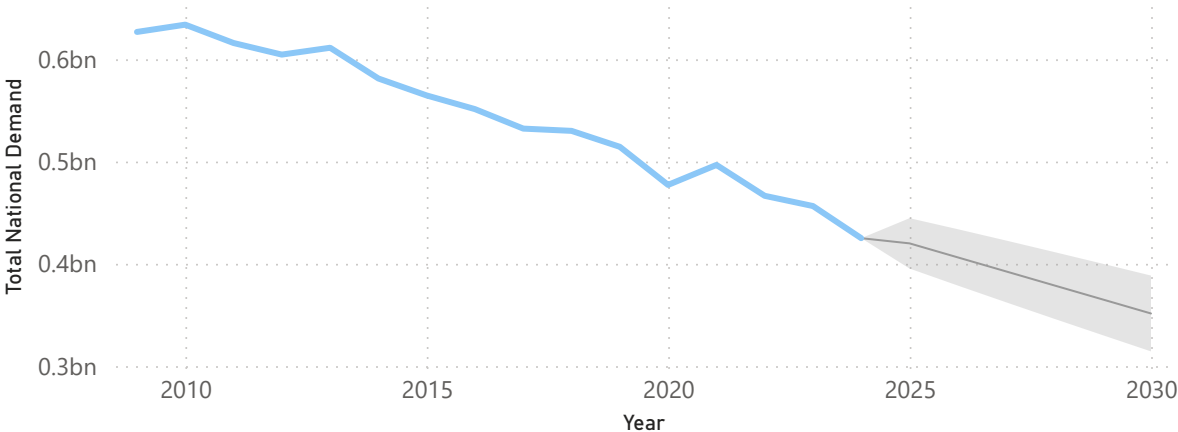
Month

All

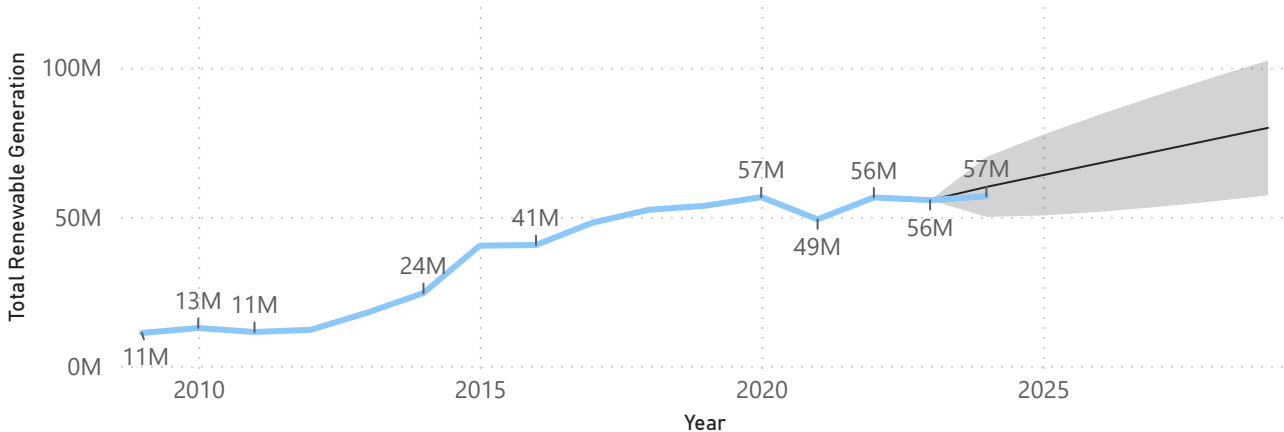
Is_holiday

All

Demand Forecast for Next 6 years



Solar & Wind Energy Trends



599M

Total Renewable Generation

8bn

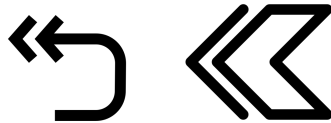
Total_Non_Renewable

6.90

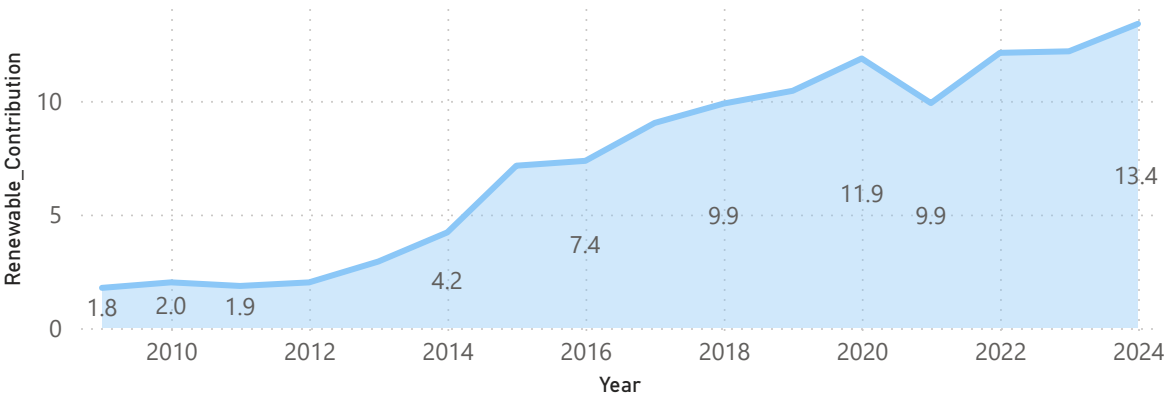
Renewable_Contribution

0.07

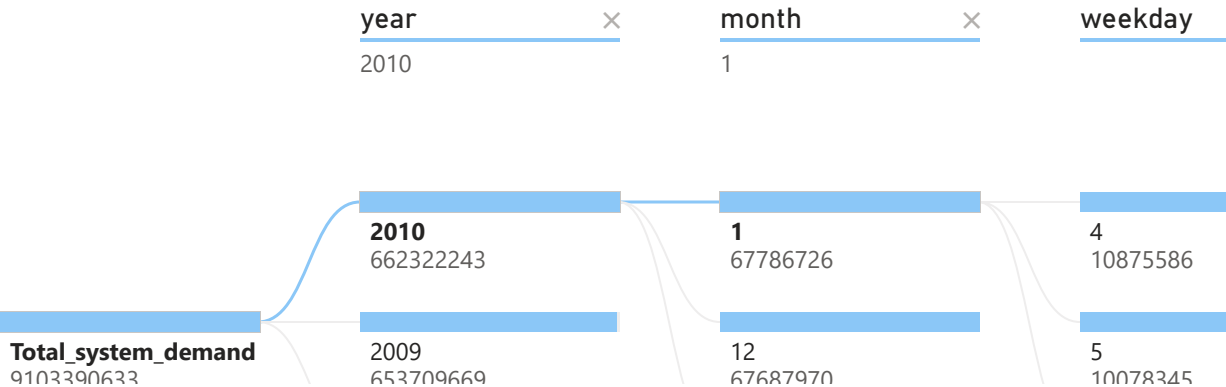
Renewable %



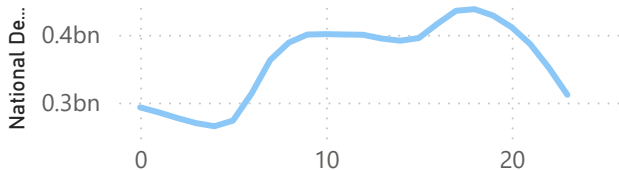
Renewable % Contribution Over Time



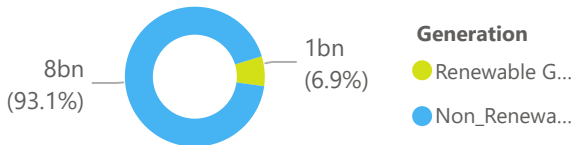
Seasonal Demand Breakdown



National Demand by hour

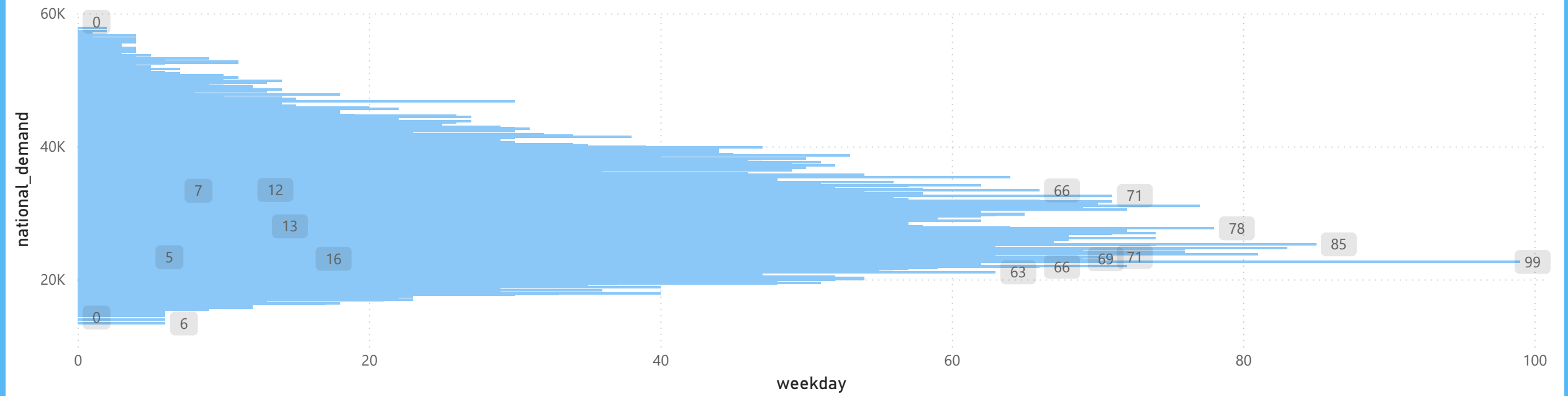


Renewable Generation and Non_Renewable





Electricity Demand Breakdown by Weekday



Monthly Demand Trends

